

while Loop Statements

```
count = 0
```

```
while (count < 9):
```

```
    print ('The count is:', count)
```

```
    count = count + 1
```

```
print ("Good bye!")
```

output

The count is: 0

The count is: 1

The count is: 2

The count is: 3

The count is: 4

The count is: 5

The count is: 6

The count is: 7

The count is: 8

Good bye!

```
count = 0
```

```
while count < 7:
```

```
    print (count, " is less than 7")
```

```
    count = count + 1
```

```
else:
```

```
print ("Loop End")
```

output

0 is less than 7

1 is less than 7

2 is less than 7

3 is less than 7

4 is less than 7

5 is less than 7

6 is less than 7

Loop End

for Loop Statements

```
range(0, 5)
```

```
[0, 1, 2, 3, 4]
```

```
for var in list(range(5)):
```

```
    print (var)
```

output

0

1

2

3

4

Break / Continue statement

1.break

```
for i in range(1, 6):
```

```
    if i == 3:
```

```
        break
```

```
    print(i)
```

output

1

2

2. continue Statement

```
for i in range(1, 6):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```

output

1

2

4

5

Pass statement

Empty Condition Block

```
x = 10
```

```
if x > 5:
```

```
    pass
```

```
else:
```

```
    print("Less than 5")
```

```
print("Done")
```

output

Done

Old String formate

```
name = "John"
```

```
print("Hello, %s!" % name)
```

output

Hello, John!

```
name = "John"
```

```
age = 23
```

```
print("%s is %d years old." % (name, age))
```

output

John is 23 years old.

New String Format {} Method

Basic format()

```
name = "John"
```

```
print("My name is {} and I am {} years old.".format(name, age))
```

output

My name is John and I am 23 years old.

Python program to do arithmetical operations

```
num1 = input("Enter first number: ")
```

```
num2 = input("Enter second number: ")
```

```
num1 = float(num1)
```

```
num2 = float(num2)
```

```
total = num1 + num2
```

```
difference = num1 - num2
```

```
product = num1 * num2
```

```
quotient = num1 / num2
```

```
print("The sum of {0} and {1} is {2}".format(num1, num2, total))
```

```
print("The subtraction of {0} and {1} is {2}".format(num1, num2, difference))
```

```
print("The multiplication of {0} and {1} is {2}".format(num1, num2, product))
```

```
print("The division of {0} and {1} is {2}".format(num1, num2, quotient))
```

output

Enter first number: 6

Enter second number: 3

The sum of 6.0 and 3.0 is 9.0

The subtraction of 6.0 and 3.0 is 3.0

The multiplication of 6.0 and 3.0 is 18.0

The division of 6.0 and 3.0 is 2.0

Python program to swap two variables

```
x = input("Enter value of x: ")
```

```
y = input("Enter value of y: ")
```

```
temp = x
```

```
x = y
```

```
y = temp
```

```
print("The value of x after swapping: {}".format(x))
```

```
print("The value of y after swapping: {}".format(y))
```

output

Enter value of x: 4

Enter value of y: 8

The value of x after swapping: 8

The value of y after swapping: 4

Python Program to Check if a Number is Odd or Even

```
num = int(input("Enter a number: "))
```

```
if num % 2 == 0:

    print("{0} is Even number".format(num))

else:

    print("{0} is Odd number".format(num))
```

output

Enter a number: 5

5 is Odd number

Python Program to check if a Number is Positive, Negative or Zero

```
num = float(input("Enter a number: "))
```

```
if num > 0:

    print("{0} is a positive number".format(num))

elif num == 0:

    print("{0} is zero".format(num))

else:

    print("{0} is a negative number".format(num))
```

output

Enter a number: 5

5.0 is a positive number

